Alexander Elzenaar

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Research interests

- Groups related to geometry and topology in low dimensions. Kleinian groups and their deformation theory; other aspects of geometric group theory, conformal geometry, and 3-manifold theory. Applications of techniques from representation theory and algebraic geometry.
- Exemplar MSC codes: 20F65, 20H10, 22E40, 30F40, 30F60, 32G15, 51F15, 57K31, 57K32.

Tertiary education

School of Mathematics, Monash University **Melbourne, Australia**
Doctoral student, geometry and topology *2024–ongoing*
Advisor: Prof. Jessica Purcell.
Expected completion: 2027.

Dept. of Mathematics, The University of Auckland **Auckland, NZ**
MSc with First Class Honours in mathematics *2021–2022*
Advisors: Dist. Prof. Gaven Martin (NZ Inst. Adv. Study., Massey Uni.) and Dr. Jeroen Schillewaert.
Thesis: *Deformation spaces of Kleinian groups* (`https://aelzenaar.github.io/msc_thesis.pdf`)

Dept. of Mathematics, The University of Auckland **Auckland, NZ**
BSc (Hons) with First Class Honours in mathematics *2020*
Advisor: Dr. Jeroen Schillewaert.
Dissertation: *Toric varieties* (`https://aelzenaar.github.io/hons/dissertation.pdf`)

School of Cultures, Languages and Linguistics, The University of Auckland **Auckland, NZ**
Certificate of Languages *awarded 2020*
Russian and Ancient Egyptian

The University of Auckland **Auckland, NZ**
BSc, major in mathematics *2017–2019*

The University of Canterbury **Christchurch, NZ**
STAR programme *2016*
Distance learning first year university mathematics for secondary school students

Employment history (research and non-academic)

Ministry of Business, Innovation, and Employment **Wellington, NZ**
Research and data analyst, Evidence & Insights branch *2023–2024*
Mathematical modelling and statistics to support ministerial offices and policy evaluators. Contributions to, and peer review of, public research output.

Max-Planck-Inst. für Math. in den Naturwissenschaften **Leipzig, Germany**
Doctoral student *2022–2023*
Non-linear algebra group. Withdrew for personal reasons.

Dept. of Mathematics, The University of Auckland **Auckland, NZ**
Research assistant *2020–2021*
- Construction and symmetry properties of spherical (t, t)-designs (funded by Dr. Shayne Waldron)
- Teaching of semester-long graduate seminar on Kleinian groups (funded by Dr. Jeroen Schillewaert)

Employment history (teaching)

School of Mathematics, Monash University **Melbourne, Australia**
Sessional teaching associate *2025–2026*
- Tutoring and marking MAT 1030 (linear algebra service course), sem. 1 and sem. 2, 2026
- Tutoring and marking MAT 1020 (calculus service course), sem. 1, 2025

Dept. of Mathematics, The University of Auckland **Auckland, NZ**
Graduate teaching assistant *2019–2022*
- Marking MATHS 208 (multivariate calculus service course), summer sem., 2022
- Tutoring & marking MATHS 332 (real analysis); tutoring MATHS 190 (misc. topics inc. cryptography, infinities, tilings, fractals), sem. 2, 2021
- Tutoring & marking MATHS 250 (calculus & linear algebra II); tutoring COMPSCI 225 (discrete mathematics for CS); marking MATHS 254 (number theory; groups; metric spaces), sem. 1, 2021
- Tutoring & marking MATHS 190 (misc. topics inc. cryptography, infinities, tilings, fractals); tutoring MATHS 120 (linear algebra); marking MATHS 340 (real & complex calculus), sem. 2, 2020
- Tutoring MATHS 162 (computational mathematics & combinatorics); assistance room tutor; marking MATHS 120 (linear algebra), MATHS 130 (calculus), COMPSCI 120 (mathematics for CS), sem. 1, 2020
- Marking MATHS 253 (calculus & linear algebra III), sem. 2, 2019

Academic works

Preprints and submitted

[1] A. Elzenaar, Quasi-Fuchsian groups and complex realisations of q-deformed real numbers, 2026. arXiv 2609.29831 [math.CV].

[2] A. Elzenaar, On rank two Kleinian groups with three parabolics, 2026. arXiv 2607.07424 [math.GR].

[3] A. Elzenaar, Expansion joints in hyperbolic manifolds, 2025, Updated Jun 2026. arXiv 2512.00879 [math.GT].

[4] A. Elzenaar, Peripheral subgroups of Kleinian groups, 2025, Updated Dec 2025. arXiv 2508.00297 [math.GT].

[5] A. Elzenaar, From disc patterns in the plane to character varieties of knot groups, 2025, Updated Aug 2025. arXiv 2503.13829 [math.GT].

[6] A. Elzenaar, Changing topological type of compression bodies through cone manifolds, 2024, Updated Mar 2026. arXiv 2411.17940 [math.GT].

[7] A. Elzenaar, G. J. Martin, and J. Schillewaert, On thin Heckoid and generalised triangle groups in $\mathrm{PSL}(2,\mathbb{C})$, NUS/IMS Scientific Reports: Computational Aspects of Thin Groups, 03 Jun 2024–14 Jun 2024, 2024. arXiv 2409.04438 [math.GT].

[8] A. Elzenaar, J. Gong, G. J. Martin, and J. Schillewaert, Bounding deformation spaces of Kleinian groups with two generators, 2024, Updated Jun 2026. arXiv 2405.15970 [math.CV].

Published

[9] A. Elzenaar, G. J. Martin, and J. Schillewaert, The combinatorics of Farey words and their traces, *Groups, Geometry, and Dynamics* **20** no. 1 (2026), 215–256. MR 5032293. Zbl 8166380. arXiv 2204.08076 [math.GT]. `https://doi.org/10.4171/GGD/832`.

[10] A. Elzenaar and S. Waldron, Putatively optimal projective spherical designs with little apparent symmetry, *Journal of Combinatorial Designs* **33** no. 6 (2025), 222–234. MR 4895104. Zbl 1562.05025. arXiv 2405.19353 [math.CO]. `https://doi.org/10.1002/jcd.21979`.

[11] A. Elzenaar, G. J. Martin, and J. Schillewaert, Concrete one complex dimensional moduli spaces of hyperbolic manifolds and orbifolds, in *2021–22 MATRIX Annals* (D. R. Wood, J. de Gier, and C. E. Praeger, eds.), Springer, 2024, pp. 31–74. MR 4807267. Zbl 1520.30060. arXiv 2204.11422 [math.GT].

[12] A. Elzenaar, G. J. Martin, and J. Schillewaert, Approximations of the Riley slice, *Expositiones Mathematicae* **41** no. 1 (2023), 20–54. MR 4557273. Zbl 1520.30060. arXiv 2111.03230 [math.GT]. https://doi.org/10.1016/j.exmath.2022.12.002.

Awards and scholarships

- Program associate fellowship, SLMath (formerly MSRI), Topological and Geometric Structures in Low Dimensions Program (Jan. to Mar. 2026)
- CARMA-MATRIX Maths Art Poster Competition, first prize in the outreach category for *Fractals that parameterise hyperbolic handlebodies* (2025)
- Research Training Program Scholarship and Stipend (Australian Govt.) (tenure 2024–2027)
- Clay Mathematics Institute Early Career Researcher Support for attendance of NZMRI Summer School on Groups and Dynamics, Nelson (2023)
- Kalman Summer Scholarship for attendance of NZMRI summer meeting on number theory, Akaroa (2022)
- Uni. of Auckland Dept. of Mathematics Student Research Conference prize (2021)
- Kalman Summer Scholarship for attendance of NZMRI summer meeting, Napier (2021)
- Uni. of Auckland Postgraduate Honours Scholarship (tenure 2020)
- Uni. of Auckland Summer Research Scholarship for *Numerical construction of spherical* (t, t)*-designs* with Dr. Shayne Waldron (2019)
- Uni. of Auckland Faculty of Arts Dean's List (2017)
- NZQA Outstanding Scholar Award, limited to top 40–60 secondary school students in New Zealand, including an Outstanding Scholarship in Classical Studies (one of 13 awarded nationally) (2016)
- Royal Society of New Zealand scholarship to attend XVI Summer Research School in Mathematics and Informatics, Blagoevgrad, Bulgaria (2016)

Selected talks

- *What does algebra have to do with knot theory?*, Oberseminar Deformationsquantisierung und Geometrie (Uni. Würzburg), 16 July 2026.
- *Towards a structure theory for rank two Kleinian groups*, Low-Dimensional Topology workshop (Mathematisches Forschungsinstitut Oberwolfach), 9 July 2026.
- *Quantifying shape and position of discreteness loci in* $\mathsf{PSL}(2, \mathbb{C})$, Séminaire (Laboratoire de Mathématiques de Reims), 23 June 2026.
- *Hands-on cone deformations of hyperbolic manifolds*, Geometry, Dynamics, and Zeta Functions (Radboud Uni., Nijmegen), 18 June 2026.
- *Quantifying shape and position of discreteness loci in* $\mathsf{PSL}(2, \mathbb{C})$, Postgraduate seminar, Topological and Geometric Structures in Low Dimensions (SLMath, Berkeley), 13 March 2026.
- *Generalised Ford domains arising from peripheral geometry*, 69th Annual Meeting of the Australian Mathematical Society (La Trobe Uni.), 9 December 2025.
- *What do knots have to do with algebra?*, 2025 NZMS Colloquium (Uni. of Waikato), 27 November 2025.
- *Discontinuous subgroups of* $\mathrm{Aut}(\mathbb{S}^2)$ *come in real-algebraic families with stable combinatorics*, 9th Australian Algebra Conference (La Trobe Uni.), 18 November 2025.

- *Cone manifolds and compression bodies*, Topology Seminar (Monash Uni.) 22 October 2025.
- *Deformations of 3-orbifold holonomy groups and applications*, Early Career Showcase in Low-Dimensional Topology, Joint Meeting of the NZMS, AustMS, and AMS (Uni. of Auckland), 11 December 2024.
- *Combinatorial structures in trace polynomials of function groups*, 8th Australian Algebra Conference (ANU, Canberra), 28 November 2024.
- *Two-bridge knots, genus two surfaces, and discrete groups with two generators*, Hodgsonfest: Geometry and topology in low dimensions (Uni. Melbourne) 12 November 2024.
- *Is* $\mathsf{PSL}(2,\mathbb{Z})$ *discrete?*, Topology Seminar (Monash Uni.) 24 July 2024.
- *The dynamic in the static: Manifolds, braids, and classical number theory*, Regiomontanus PhD Seminar (Uni. Leipzig), 10 May 2023.
- *What is a Kleinian group?*, Australian Postgraduate Algebra Colloquium, 21 September 2022.
- *Pictures of hyperbolic spaces*, Discrete Mathematics and Geometry Seminar (TU Berlin), 4 May 2022.
- *Strange circles: The Riley slice of quasi-Fuchsian space*, Seminar on Nonlinear Algebra (MPI MiS), 27 April 2022.
- *Approximating the Riley slice exterior*, Matrix Inst. workshop on Groups and Geometries, 2 December 2021.
- *Real varieties of spherical designs*, Algebra and Combinatorics Seminar (Uni. of Auckland), 1 April 2021.

Professional service

- Refereeing for scholarly publications.
- Reviewing for zbMATH Open and Mathematical Reviews.

Selected events organised

- Minicourse on knot theory and geometry (Uni. Auckland), including guest lectures by Lavender Marshall and Josh Lehman (July 2023).
- Reproducibility in Computer Algebra event (MPI-MiS/TU Berlin), co-organised with Christiane Görgen and Lars Kastner (August 2022).
- A Day of Geometry and Lorentzian Polynomials (MPI-MiS), single day of short informal talks in preparation for Göran Gustafsson Symposium at the Institut Mittag-Leffler (May 2022).
- Seminar on Kleinian groups (Uni. Auckland), weekly lectures covering approximately the first half of *Kleinian Groups* by B. Maskit along with some sections of Thurston's lecture notes and the book *Hyperbolic Manifolds and Discrete Groups* by M. Kapovich. Funded by Jeroen Schillewaert through a Research Assistantship (2021).
- Reading group on algebraic geometry (Uni. Auckland), jointly organised with Oliver Li (2020–2021).

Professional affiliations

- New Zealand Mathematical Society
- Australian Mathematical Society

References

Teaching and research reference letters available on request.

[September 25, 2026]